

AgPlane: Letting Agents Harness Their Own System-Level Behavior with eBPF

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AI agents are widely used for coding, DevOps, and enterprise workflows[9, 13, 22, 24, 25], increasingly in production deployments that run on real infrastructure, handle sensitive data, and interact with CI/CD pipelines. Besides the LLM, agents require AI agent harnesses: software around the model that improves agent performance while enforcing behavioral constraints for safety, security, and compliance [2].

A major component of such a harness is a policy engine [20]: the part that observes and enforces constraints over the agent’s concrete actions. In practice, projects encode many such constraints as behavioral policies in instruction files (CLAUDE.md, AGENTS.md) [4, 15], so harnesses can provide these instructions to the model directly. Because LLMs comply with instructions probabilistically [12, 14, 16] and may violate them through planning errors, prompt-injection drift, and opaque tool or script behavior [9, 11, 25], harnesses also require deterministic policy engines to enforce compliance at runtime [8, 17, 21].

Many of these policies constrain system-level actions, yet existing policy engines leave a *semantic gap*: *AI agent intent and directives* are expressed in natural language, but enforcement must decide over *system actions*. Our empirical study of instruction-files in 64 highly popular projects finds that 64% statements are behavioral directives, and 83% involving system-level behavior: commands executed, files accessed, and network connections made. Existing approaches cannot close this gap. Tool-call-based policy systems [7, 8, 18, 21, 23] in agent runtimes such as Claude Code and Codex, whether LLM guardrails or regex-like checkers, lose track of indirect system actions when agents shell out or spawn subprocesses: the same `git push` can be reached through a `bash -c` wrapper, or a compiled binary created in a previous session; and at the repository level, 81% projects contain cross-event directives that single tool-level interception cannot enforce (e.g., “run tests before commit”).

OS-level enforcement with static policy is also insufficient: most directives reference project or task context that static rules cannot resolve (e.g., “never modify upstream source,” where *upstream source* must be resolved from the repository; “do not update dependencies unless requested”; “the reviewer sub-agent only writes to its project directory”). We

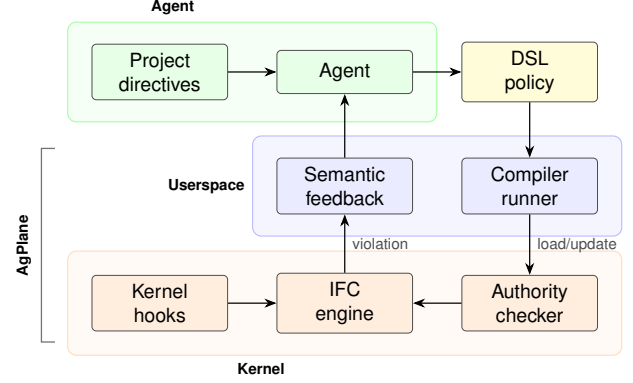


Figure 1. AgPlane architecture: policy generation, compilation, kernel enforcement, and feedback loop.

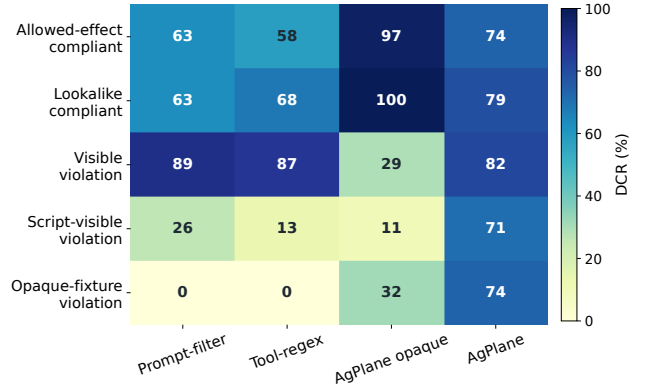


Figure 2. RQ1: Decision Compliance Rate by trace family and system. AgPlane’s advantage concentrates on indirect execution paths (script and hidden families) that tool-call baselines cannot observe.

find that 74% system-level directives require context absent from low-level OS events. OS-level enforcement systems and sandboxes for traditional software [3, 5, 6, 19] can observe all system actions but expect pre-defined static policy and return only system events without semantic context, leaving the agent unable to follow its behavioral directives.

We argue that an agent-harness policy engine should be programmable by agents and take effect at OS level: it should let agents use live project and task context to enforce actions and adapt at runtime. Our key observation is that, while

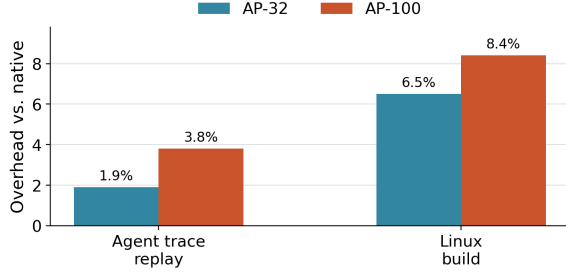


Figure 3. RQ2: End-to-end overhead normalized to native execution, for agent trace replay and Linux kernel build.

safety constraints must come from higher authority, the required context is mainly within agents closest to the task: they already read the repository, interpret the current task, and resolve abstract references into concrete commands and paths before enforcement can begin. This makes agents the natural producer of concrete policy from directives, also reflecting the fact that instruction files are increasingly maintained by agents [1, 10].

Realizing this creates three challenges. First, policy specification must be high-level enough for agents to easily generate from natural language with minimal token consumption, yet concrete enough to compile to deterministic kernel checks. Second, enforcement must track policy-relevant state across full system without imposing prohibitive per-syscall overhead. Third, since many rules are about safety and security, programmability must not become an authority bypass. Unlike traditional sandboxes threat model where a trusted administrator authors policy, here agents are both author and constrained subject, so they need an interface to add constraints without weakening inherited policy set by trusted administrators or parent agents or introducing safety risks.

AgPlane is a programmable OS-level policy enforcement system for AI agent harnesses. Agents express policies in a compact DSL generated from natural-language; AgPlane compiles them into labeled information-flow rules and loads them into an eBPF enforcement engine across processes, files, and network kernel hooks. When matching a rule, AgPlane returns semantic feedback: which rule matched, why, and what to do next. Each rule independently selects one of three effects: notify (observe), block (pre-operation denial), or kill (process termination). A layered authority model ensures that higher-authority rules (e.g., global “do not leak secrets”) cannot be weakened by agent-authored policy. On our decision-compliance benchmark, AgPlane detects 2.4× more violations than prompt-filter or tool-regex hooks by covering indirect execution paths that tool-call interception cannot, while adding 1.9% end-to-end overhead on agent workloads and 8.4% on I/O-intensive kernel builds. On a third-party safety benchmark of 361 workplace and personal-assistant

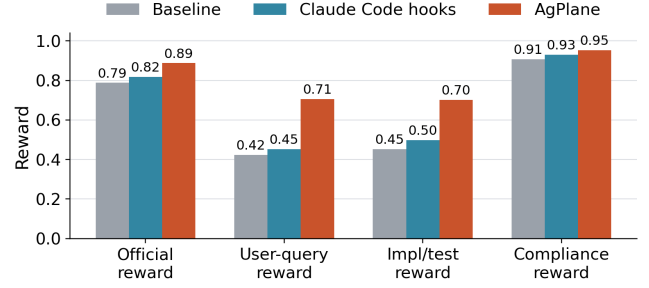


Figure 4. RQ3: OctoBench 20-task subset reward breakdown by system.

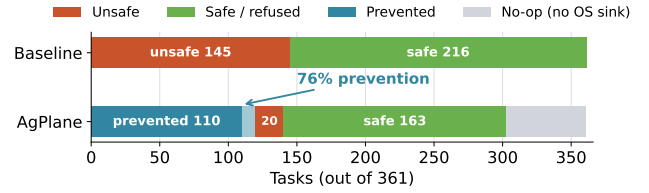


Figure 5. RQ4: Outcome distribution on 361 OpenAgentSafety tasks. AgPlane reduces baseline-unsafe outcomes from 145 to 20 (76% prevention rate).

tasks, AgPlane prevents 76% of baseline-unsafe behaviors using automatically generated policies.

We make three contributions:

1. An **empirical study** of 64 projects that characterizes these gaps.
2. **AgPlane**, a programmable OS-level policy enforcement system that addresses them.
3. An **evaluation** on our decision-compliance benchmark building on empirical study, external coding-task and safety benchmarks covering workplace and personal-assistant tasks.

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